DIGITAL SAT MATH PRACTICE

Radical and Rational Exponents

20 Questions | ~20% Grid Ins | Difficulty Labeled

TOPIC: RADICAL AND RATIONAL EXPONENTS

Question 1

[Easy]

Which of the following is equivalent to √($49x^4$)?

- A) $7x^2$
- B) $7x^4$
- C) $49x^2$
- D) $7x$

Question 2

[Easy]

What is the value of $27^{1/3}$?

- A) 3
- B) 9
- C) 81
- D) 1/3

Question 3

[Easy]

Which expression is equivalent to $x^{3/4}$?

- A) ■√(x^3)
- B) ^{3}√(x■)
- C) (x^3)/4
- D) 3√x

Question 4

[Easy]

What is the value of $16^{3/4}$?

A) 4
B) 8
C) 12
D) 64

Question 5

[Medium]

If $\sqrt{(5k + 11)} = k + 1$, what is the value of k?

A) –2 only
B) 5 only
C) –2 and 5
D) No solution

Question 6

[Medium]

Which expression is equivalent to $(8m^6)^{2/3}$?

A) $4m^4$
B) $4m^9$
C) $2m^4$
D) $2m^9$

Question 7

[Medium]

If $x^{1/2} = 5$, what is the value of $x^{3/2}$?

A) 15
B) 25
C) 75
D) 125

Question 8

[Medium]

Simplify: $(x^{2/5})(x^{3/5})$.

- A) $x^{6/25}$
- B) x^{1}
- C) $x^{6/5}$
- D) $x^{5/10}$

Question 9

[Medium]

The expression $\sqrt[3]{(64y^9)}$ is equivalent to which of the following?

- A) $4y^3$
- B) $8y^3$
- C) $4y^6$
- D) $8y^6$

Question 10

[Medium]

What is the value of $32^{2/5}$?

Student Produced Response (Grid In)

Question 11

[Medium]

Which expression is equivalent to $x^{-1/2}$?

- A) $-\sqrt{x}$
- B) $1/\sqrt{x}$
- C) $\sqrt{(-1/x)}$
- D) $-x/2$

Question 12

[Hard]

If $3^{2x} = 81$ and $2^{y} = 16$, what is the value of $x + y$?

- A) 4
- B) 6
- C) 8
- D) 10

Question 13

[Hard]

If $(x^a)^3 \cdot x^4 = x^{19}$, what is the value of a?

Student Produced Response (Grid In)

Question 14

[Hard]

The expression $(4x^6y^{-2})^{1/2}$ is equivalent to which of the following?

- A) $2x^3/y$
- B) $2x^3y$
- C) $4x^3/y$
- D) $2x^{12}/y$

Question 15

[Hard]

Which of the following is equivalent to $\sqrt{(50x^3)}$ in simplest radical form?

- A) $5x\sqrt{(2x)}$
- B) $25x\sqrt{(2x)}$
- C) $5x^2\sqrt{2}$
- D) $10x\sqrt{x}$

Question 16

[Hard]

If $f(x) = 2x^{3/2}$ and $g(x) = x^{1/3}$, what is f(g(64))?

- A) 8
- B) 16
- C) 32
- D) 64

Question 17

[Hard]

The expression $(x^4y^{-6})/(x^{-2}y^3)$ can be written as x^ay^b. What is the value of $a - b$?

Student Produced Response (Grid In)

Question 18

[Hard]

If $p^{1/3} + p^{1/3} + p^{1/3} = 12$, what is the value of p?

- A) 4
- B) 36
- C) 48
- D) 64

Question 19

[Hard]

If $5^x \cdot 5^{x+1} = 5^9$, what is the value of x?

Student Produced Response (Grid In)

Question 20

[Hard]

The expression (√[3]{a^2b})(√[3]{ab^5}) can be rewritten in the form $a^m b^n$. What is the value of m + n?

- A) 2
- B) 3
- C) 7/3
- D) 8/3